KAPASA MAKASA UNIVERSITY

ICT DEPARTMENT

CYS 110 TEST 1

24th February 2023

INSTRUCTIONS

Do not turn this page until you are told to do so

This Question paper contains 5 Questions, attempt all questions.

Write your name, student ID and Program clearly on the answer sheet.

Time allowed is 2 hours only

Only calculators are allowed in the Exam room.

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[QUESTION 1]

A. Using a Caesar cipher with S = 5 decode the received message **YMNX YJXY NX NSYJWJXYNSL.** **[5 Marks]**

B. Using a Caesars cipher with S = 3 and S = 7, Encrypt the message WELCOME TO CRYPTOGRAPHY using a C1C2C2C1C2 sequence **[5 Marks]**

C. The centurion who was supposed to inform you of s was killed en route, but you have received the message MXX SMGX UE PUHUPQP in a Caesar cipher. Find the value of s and decode the message. Hint: *Use Brute Force* **[10 Marks]**

[Total 20 Marks]

[QUESTION 2]

A. You receive a message that was encoded using a block encoding scheme with the encoding matrix, Decode the cipher text MXOSHI

$M = \begin{bmatrix} 3 & 2 \\ 7 & 5 \end{bmatrix}$ $M' = \begin{bmatrix} 5 & 24 \\ 19 & 3 \end{bmatrix}$

Prove that $M \times M' = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ **[15 Marks]**

B. The DES algorithm combines two bit strings by applying the XOR operator on each pair of corresponding bits. Compute the 6-bit string that results from 100111 ⊕ 110101 **[5 Marks]**

[Total 20 Marks]

[QUESTION 3]

A. A virus attacks a single user's computer and within one hour embeds itself in 50 e-mail attachment files sent out to other users. By the end of the hour, 10% of these have been opened and have infected their host machines. If this process continues, how many machines will be infected at the end of 5 hours? Can you find a formula for the number of machines infected after n hours? **[5 Marks]**

B. Using the RSA encryption algorithm, let p = 3 and q =5. Then n = 15 and m = 8. Let e = 11. **[15 Marks]**

i. Compute d.

ii. Find the code for 3.

iii. Decode your answer to part (ii) to retrieve the 3.

[Total 20 Marks]

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[QUESTION 4]

A. You receive a message "17" that was sent using the RSA public encryption key (21, 5) of the example in this section. Decode this to find the original numeric message. **[10 Marks]**

B. If **DREA** is the Key, use columnar transposition to decrypt the message **SEAZITFQIMRITIOU** **[10 Marks]**

[Total 20 Marks]

[QUESTION 5]

A. State and explain any five active attacks that can be conducted by a hacker and cause a compromise on the confidentiality of the clear message m. **[10 Marks]**

B. Explain the main differences between RC4 and RC5 stating their advantages and disadvantages **[10 Marks]**

[Total 20 Marks]

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CYS 110

TEST 1 (23)

QUESTION 1

A. Using a Caesar cipher with S = 5, decode the received message YMNX YJXY NX NSYJWJXYNSL.

ANSWER:

Plain text : this test is interesting
Cipher text : YMNX YJXY NX NSYJWJXYNSL

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z	plain text
F G H I J K L M N O P Q R S T U V W X Y Z A B C D E	Cipher text

B. Using a Caesars cipher with S = 3 and S = 7, Encript the message WELCOME TO CRYPTOGRAPHY using a $C_1C_2C_2C_1C_2$ Sequence.

ANSWER:

	C_1 C_2 C_2 C_1 C_2	C_2 C_1	C_2 C_1 C_2 C_2 C_1 C_2 C_1 C_2 C_2 C_1
Plain text :	WELCOME	TO	CRYPTOGRAPHY
Cipher text :	ZLSFVPL	AR	JUFNWVIYHSOB

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z	plain text
D E F G H I J K L M N O P Q R S T U V W X Y Z A B C	S = 3 (C_1)
H I J K L M N O P Q R S T U V W X Y Z A B C D E F G	S = 7 (C_2)

C. The centurion who was supposed to inform you of S was killed en route, but you have received the message MXX SMGX UE PUHUPQP in a Caesar cipher. Find the value of S and decode the message.

Hint: Use Brute Force

ANSWER:

Plain text: ALL GAUL IS DIVIDED
Cipher text: MXX SMGX UE PUHUPQP

MXX SMGX UE PUHUPQP
is
if
as
or

Cipher text A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
plain text O P Q R S T U V W X Y Z A B C D E F G H I J K L M N

if S = 18, S = A ∴ the key is 14.
if S = 18 - 4 S = E
R - D
= 14

QUESTION 2

A. You receive a message that was encoded using a block encoding scheme with the encoding matrix, Decode the cipher text MXOSHI.

$$M = \begin{pmatrix} 3 & 2 \\ 7 & 5 \end{pmatrix} \qquad M' = \begin{pmatrix} 5 & 24 \\ 19 & 3 \end{pmatrix}$$

Prove that $M \times M' = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

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ANSWER:

0	1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L	M
1	2	3	4	5	6	7	8	9	10	11	12	13
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25
14	15	16	17	18	19	20	21	22	23	24	25	26

Plain text: a e t a c k

Cipher text: M X O S H I

$$(M \quad X)\begin{pmatrix}5 & 24\\ 19 & 3\end{pmatrix}$$

$$(13 \quad 24)\begin{pmatrix}5 & 24\\ 19 & 3\end{pmatrix}$$

$$[(13\times5)+(24\times19) \quad (13\times24)+(24\times3)]$$

$$[(65+456) \quad (312+72)]$$

$$(521 \quad 384)$$

$\frac{521}{26} = 20.03$ $\qquad$ $\frac{384}{26} = 14.76$

$20 \times 26 = 520$ $\qquad$ $14 \times 26 = 364$

$521 - 520 = 1$ $\qquad$ $384 - 364 = 20$

$S(1) = A$ $\qquad$ $S(20) = T$

$$(O \quad S)\begin{pmatrix}5 & 24\\ 19 & 3\end{pmatrix}$$

$$(15 \quad 19)\begin{pmatrix}5 & 24\\ 19 & 3\end{pmatrix}$$

$$[(15\times5)+(19\times19) \quad (15\times24)+(19\times3)]$$

$$[(75+361) \quad (360+57)]$$

$$(436 \quad 417)$$

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$\frac{436}{26} = 16.76$

$16 \times 26 = 416$

$436 - 416 = 20$

$S(20) = T$

$\frac{417}{26} = 16.03$

$16 \times 26 = 416$

$417 - 416 = 1$

$S(1) = A$

$(H \quad I)\begin{pmatrix} 5 & 24 \\ 19 & 3 \end{pmatrix}$

$(8 \quad 9)\begin{pmatrix} 5 & 24 \\ 19 & 3 \end{pmatrix}$

$[(8 \times 5) + (9 \times 19) \quad (8 \times 24) + (9 \times 3)]$

$[(40 + 171) \quad (192 + 27)]$

$(211 \quad 219)$

$\frac{211}{26} = 8.11$

$8 \times 26 = 208$

$211 - 208 = 3$

$S(3) = C$

$\frac{219}{26} = 8.42$

$8 \times 26 = 208$

$219 - 208 = 11$

$S(11) = K$

$M \times M' = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$\begin{pmatrix} 3 & 2 \\ 7 & 5 \end{pmatrix}\begin{pmatrix} 5 & 24 \\ 19 & 3 \end{pmatrix}$

$\begin{bmatrix} (3 \times 5) + (2 \times 19) & (3 \times 24) + (2 \times 3) \\ (7 \times 5) + (5 \times 19) & (7 \times 24) + (5 \times 3) \end{bmatrix}$

$\begin{bmatrix} (15 + 38) & (72 + 6) \\ (35 + 95) & (168 + 15) \end{bmatrix}$

$\begin{pmatrix} 53 & 78 \\ 130 & 183 \end{pmatrix}$

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$\frac{53}{26} = 2.03$ $\frac{78}{26}$, if you round up (wrap around)

$2 \times 26 = 52$ 78 you reach at 0.

$53 - 52 = 1$ $\frac{78}{26} = 3$, $3 \times 26 = 78$

$78 - 78 = 0$

$\frac{130}{26} =$, if you wrap around 130, you rod up at 0.

$\frac{183}{26}$ $\frac{130}{26} = 5$ $\frac{183}{26} = 7.03$

$5 \times 26 = 130$ $7 \times 26 = 182$

$130 - 130 = 0$ $183 - 182 = 1$

$$\therefore \begin{pmatrix} 3 & 2 \\ 7 & 5 \end{pmatrix} \begin{pmatrix} 5 & 24 \\ 19 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$M \times M' = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

B. The DES algorithm combines two bit strings by applying the XOR operator on each pair of corresponding bits. Compute the 6-bit string that results from $100111 \oplus 110101$

ASWER :

NOTE: Same characters (e.g $1 \oplus 1 = 0$) and different characters ($1 \oplus 0 = 1$).

$\therefore$ 100111
$\oplus$ 110101
010010

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QUESTION 3

A. A virus attacks a single user's computer and within one hour embeds itself in 50 e-mail attachment files sent out to other users. By the end of the hour, 10% of those have been opened and have infected their host machines. If this process continues, how many machines will be infected at the end of 5 hours? Can you find a formula for the number of machines infected after n hours?

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8. Using the RSA encryption algorithm, let $p = 3$ and $q = 5$. Then $n = 15$ and $m = 8$. Let $e = 11$.

(i) Compute d.

ANSWER:

Prime numbers less than $n = 15$

1, 3, 5, 7, 11, 13

$\frac{ed - 1}{m}$

$\frac{ed - 1}{m}$ must give us a remainder of 0.

$\frac{(11 \times 3) - 1}{8}$

$\frac{33 - 1}{8}$

$\frac{32}{8} = 4, \; R = 0$ ∴ d is 3.

(ii) Find the code for 3.

ANSWER:

$C = m^e \bmod n$

$= 3^{11} \bmod 15$

$= 177147 \bmod 15$

$= \frac{177147}{15}$

$= 11809.8$

$= 11809 \times 15$

$= 177135$

$= 177147 - 177135$

$= 12$

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(iii) Decode your answer to part (ii) to retrieve the 3.

ANSWER:

$m = c^d \bmod n$

$= 12^3 \bmod 15$

$= 1728 \bmod 15$

$= \frac{1728}{15}$

$= 115.2$

$= 15 \times 115$

$= 1725$

$= 1728 - 1725$

$= 3$

QUESTION 4

A. You receive a message "17" that was sent using the RSA public encryption key (21, 5) of the example in this section. Decode this to find the original numeric message.

ANSWER:

1	2	3	4	5	6	7	8	9	10	11	12	13
A	B	C	D	(E)	F	G	H	I	J	K	L	M

N	O	P	(Q)	R	S	T	U	V	W	X	Y	Z
14	15	16	17	18	19	20	21	22	23	24	25	26

$(n, e) = (21, 5)$

$n = p \times q \quad \therefore p = 3$ and $q = 7$

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$m = (p-1) \times (q-1)$
$= (3-1) \times (7-1)$
$= 2 \times 6$
$= 12$

$\frac{ed-1}{m}$

$\frac{(5 \times 5) - 1}{12}$

$\frac{25-1}{12}$

$\frac{24}{12} = 2, R = 0$

$\therefore d = 5$

Private key $(n, d) = (21, 5)$

$m = c^d \bmod n$
$= 17^5 \bmod 21$
$= 1419857 \bmod 21$
$= \frac{1419857}{21}$
$= 67612.23$
$= 21 \times 67612$
$= 1419852$
$= 1419857 - 1419852$
$= 5$

B. If DREA is the key, use columnar transposition to decrypt the message SEAZITFQIMRITIOU.

ANSWER:

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1			2	3													4								

DREA has 4 characters,

Therefore, | SEAZ | ITFQ | IMRI | TIOU |
A D E R

D	R	E	A
i	t	i	s
t	i	m	e
F	o	r	a
q	u	i	z

∴ SEAZITFQIMRITIOU becomes "it is time for a quiz" when decrypted by DREA.